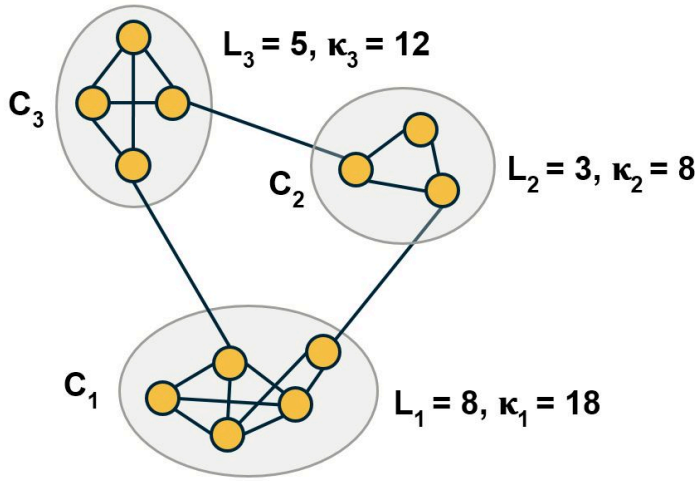


L7: Modularity Metric- Derivation



We can now derive a more useful formula for the modularity metric, starting from the definition we gave earlier:

$$M = \frac{1}{2L} \sum_{c=1}^C \sum_{(i,j) \in C_c} (A_{i,j} - \frac{k_i k_j}{2L})$$

The first term can be re-written as:

$$\frac{1}{2L} \sum_{c=1}^C \sum_{(i,j) \in C_c} A_{i,j} = \sum_{c=1}^C \frac{L_c}{L}$$

where L_c is the number of internal edges in community C_c :

$$\frac{1}{2} \sum_{(i,j) \in C_c} A_{i,j}$$

The second term can be re-written as:

$$\frac{1}{2L} \sum_{c=1}^C \sum_{(i,j) \in C_c} \frac{k_i k_j}{2L} = \frac{1}{4L^2} \sum_{c=1}^C \sum_{(i,j) \in C_c} k_i k_j = \sum_{c=1}^C \left(\frac{\kappa_c}{2L} \right)^2$$

where κ_c is the total number of stubs at the nodes of community c , i.e.,

$$\kappa_c = \sum_{i \in C_c} k_i$$

If the last part of this equation is not clear, note that:

$$\kappa_c^2 = \left(\sum_{i \in C_c} k_i \right) \left(\sum_{j \in C_c} k_j \right) = \sum_{(i,j) \in C_c} (k_i k_j)$$

Substituting these two terms back to the original modularity definition, we get an equivalent expression for the modularity metric:

$$M = \sum_{c=1}^C \left(\frac{L_c}{L} - \left(\frac{\kappa_c}{2L} \right)^2 \right)$$

This expression is quite useful because it expresses the modularity of a given community assignment as a summation, across all communities, of the following difference:

the fraction of network edges that are internal to community C_c MINUS the squared fraction of total edge stubs that belong to that community.

Food For Thought

Use this modularity formula to calculate the modularity of each of the following partitions: a) all nodes are in the same community, b) each node is in a community by itself, c) each community includes nodes that are not connected with each other, d) a partition in which there are no inter-community edges.